

SARA ELSHARKAWY

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EDUCATION

Master of Animal Biomedical Sciences, Veterinary Public Health and Epidemiology (Aug 2022 – Jun 2024)
School of Veterinary Medicine and Biomedicine, IPB University, Bogor, Indonesia.

- Thesis Title: "Analysis of beta-lactam antibiotic resistance genes in *Escherichia coli* isolated from dairy cattle manure"
- GPA: 4.0/4.0
- Awarded University Consortium (UC) Student Thesis Grant for Research Activities by SEARCA (2023)

Bachelor Of Veterinary Medical Science (Sep 2011 – Jun 2016)

Faculty of Veterinary Medicine, University of Sadat City, Menoufia, Egypt
Final Grade: Very Good with Honor

RESEARCH EXPERIENCE

First Stage Researcher (R1) (Oct 2023 - Dec 2023)

Laboratory of Quality Control and Certification of Animal Products (BPMSPH), Bogor, Indonesia.

- Gained an internship for three months
- Conducted molecular characterization of *E. coli* isolates using Real-time.
- Analyzed resistance patterns using R software for statistical interpretation of microbiological data
- Documented research findings in peer-reviewed publication in German Journal of Veterinary Research (2024)

Research Assistant (Jan 2023 – Mar 2024)

School of Veterinary Medicine and Biomedicine, IPB University, Bogor, Indonesia.

- Investigated molecular mechanisms of antibiotic resistance in bacterial pathogens with direct relevance to bacterial stress response pathways
- Applied molecular biology techniques including DNA extraction and PCR amplification for resistance gene detection
- Collaborated with interdisciplinary research team to analyze resistance patterns in agricultural settings

Research Assistant (July 2020 – July 2021)

Faculty of Veterinary Medicine, University of Sadat City, Menoufia, Egypt.

- Performed detailed ultrasonographic imaging and anatomical description of camel distal limb
- Conducted comprehensive measurements and documentation of tendon and ligament structures in camel distal limb to establish baseline anatomical references for clinical diagnostics.
- Collaborated with veterinary anatomists and clinicians to correlate ultrasonographic findings with anatomical structures for accurate diagnostic reference

TEACHING EXPERIENCE

Volunteer English Teacher for Visually Impaired Students (2021 – Present)

Resala Charity Organization, Shebin Elkom, Egypt

- Taught English to visually impaired students, adapting teaching methods to meet individual learning needs using tactile tools and audio-based resources.
- Recorded audiobooks, designed for visually impaired children.

Independent Contractor - Various Primary Schools, Online & Offline Platforms

- Delivered customized English language instruction to primary school students (ages 6-12) through both in-person and virtual sessions.
- Coordinated with multiple schools to schedule sessions and provide progress reports.
- Maintained professional relationships with various educational institutions as an independent contractor.

WORK EXPERIENCE

Owner & Lead Veterinarian

Animal Planet Veterinary Clinic

 2024 – Now  Shebin Elkom, Egypt

- Founded a veterinary clinic in my hometown, providing comprehensive medical care for companion animals and livestock.
- Conduct clinical examinations, diagnose diseases, and perform medical treatments.
- Provide preventive healthcare services including vaccination programs, deworming, and herd health management.
- Manage day-to-day clinic operations including patient flow, record keeping, and client communication.

Veterinarian

VetHouse Veterinary Clinic

 2019 – 2021  Shebin Elkom, Egypt

Pets Corner Veterinary Clinic

 2017 – 2019  Shebin Elkom, Egypt

- Performed minor surgical procedures on small animals.
- Regularly visit poultry farms to assess flock health, implement biosecurity measures, and provide on-site consultations to farmers.
- Conduct post-mortem examinations on poultry to diagnose causes of death and identify disease outbreaks.

PUBLICATIONS

1. Kusuma, N., **Elsharkawy, S.**, Tiuria, R., & Latif, H. (2026). Deteksi Gen Resistansi Antibiotik Golongan Tetrasiklin pada Escherichia coli yang Diisolasi dari Kotoran Sapi Perah di Kota Bogor [Detection of tetracycline resistance genes in Escherichia coli isolated from dairy cattle feces in Bogor City]. *Jurnal Veteriner dan Biomedis*, 4(1), 21–27. <https://doi.org/10.29244/jvetbiomed.4.1.21-27>
2. Raihan, M. A., **Elsharkawy, S.**, Indrawati, A., & Latif, H. (2026). Detection of quinolone antibiotic resistance genes in Escherichia coli isolated from dairy cattle feces. *Current Biomedicine*, 4(1), 1-7. <https://doi.org/10.29244/currbiomed.4.1.1>
3. **Elsharkawy, S.**, Latif, H., Purnawarman, T., & Rahayu, P. (2025). Prevalence of Escherichia Coli O157 strain isolated from dairy cattle manure in Bogor, Indonesia. *Revista De Ciências Agroveterinárias*, 24(3), 575–593. <https://doi.org/10.5965/223811712432025575>
4. **Elsharkawy, S.A.**, Latif, H., Purnawarman, T., & Rahayu, P. (2024). Analysis of beta-lactam antibiotic resistance genes in Escherichia coli isolated from dairy cattle manure in Bogor, Indonesia. *German Journal of Veterinary Research*, 4(3), 16-26. <https://doi.org/10.51585/gjvr.2024.3.0094>
 - First-author publication directly relevant to *E. coli* and antibiotic resistance
 - Demonstrated molecular characterization of *E. coli* resistance patterns using qPCR techniques
5. Gadallah, S.M., Sharshar, A.M., **Elsharkawy, S.A.**, & Fadel, M.S. (2023). Ultrasonographic description of tendons and ligaments at the palmar (plantar) aspect of the distal limb in the one humped camel (Camelus dromedarius). *Australian Veterinary Journal*, 101, 397-408. <https://doi.org/10.1111/avj.13268>

AWARDS AND HONORS

1. **Best Presenter award for 7th international Hybrid short course on Exotic Animal Nutrition and Feed Technology**, IPB University, Bogor, Indonesia. (Aug 2023)
2. **University Consortium (UC) Student Thesis Grant for Research activities** (Aug 2023)
Awarded 15000000 IDR to support my research on “*Analysis of beta-lactam antibiotic resistance genes in Escherichia coli isolated from dairy cattle manure in Bogor, Indonesia*”. Offered by the Southeast Asian Regional Center for Graduate Study and Research in Agriculture (SEARCA)
3. **First Winner award for Student Competition of Nutrition and Feed Technology (NuFet)**, IPB University, Bogor, Indonesia. (Aug 2022)
4. **Third Winner award for Best Student category in E-Summer Course Business Innovation and Management (BIM)**, IPB University, Bogor, Indonesia. (Aug 2022)
5. **Third Winner award for Best Three-Minute Thesis (3MT) in International Competition E-Summer Course Business Innovation and Management (BIM)**, IPB University, Bogor, Indonesia. (Aug 2022)
6. **KNB scholarship** (June 2021)
Awarded the **from the Indonesian government**, to pursue my Master degree in one of the top three universities in Indonesia IPB.

WORKSHOPS AND TRAINING

Basic Lab 12 in Scientific Research Egypt Scholars Inc.	(May 2026)
Master Microsoft Power BI Alison EMPOWER YOURSELF	(Aug 2025)
McKinsey.org Forward Program McKinsey.org	(Jul 2025)
AI-Augmented Professional Development Skills in the Digital Age African Leadership X	(Nov 2024)
Raising Awareness on Responsible Use of Antibiotics in Livestock FAO Learning Academy	(Dec 2024)
Scientific Publication Training IPB University	(Jun 2023)
Tropical Agriculture Environment toward Sustainability IPB University	(Jul 2022)
Sustainable Business Development (SBD) IPB University	(Jul 2022)
Communication and Interpersonal Skills at Work University of Leeds and Institute of Coding	(Oct 2020)

LANGUAGES

- **Arabic:** Native
- **English:** Fluent (C1)
- **Indonesian:** Intermediate

SKILLS

Research & Laboratory Techniques

- Quantitative research design
- Molecular microbiology techniques (qPCR, DNA extraction, bacterial culture)
- Antimicrobial susceptibility testing

Data Analysis & Software Tools

- Statistical analysis using SPSS and R
- Interpretation of clinical and environmental datasets
- Microsoft Office applications for data organization and basic analysis (Excel), scientific writing and reporting (Word), and scientific presentations (PowerPoint)

Scientific Communication

- Academic writing and publication

- Conference presentations and scientific reporting

Core Competencies

- Critical thinking and problem-solving
- Collaboration in interdisciplinary research teams
- Time management and attention to detail

REFERENCES

Available Upon Request